$$x - y - z + u = 0$$
$$2x + 2z = 8$$
$$-y - 2z = -8$$
$$3x - 3y - 2z + 4u = 7$$

$$\left[\begin{array}{cccc|c} 1 & -1 & -1 & 1 & 0 \\ 2 & 0 & 2 & 0 & 8 \\ 0 & -1 & -2 & 0 & -8 \\ 3 & -3 & -2 & 4 & 7 \end{array}\right]$$

A

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A = \left[\begin{array}{cccc|c} 1 & -1 & -1 & 1 & 0 \\ 0 & 2 & 4 & -2 & 8 \\ 0 & -1 & -2 & 0 & -8 \\ 3 & -3 & -2 & 4 & 7 \end{array}\right]$$

$$f_{41} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -3 & 0 & 0 & 1 \end{bmatrix}$$

$$A = \left[\begin{array}{cccc|c} 1 & -1 & -1 & 1 & 0 \\ 0 & 2 & 4 & -2 & 8 \\ 0 & -1 & -2 & 0 & -8 \\ 0 & 0 & 1 & 1 & 7 \end{array}\right]$$

$(-1 \times -3) = 3$

$3 - 3 = 0$

$$E_{23} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & \frac{1}{2} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

↑ 1st row can't be used for eliminating 0 below 2nd pivot

$$A = \left[\begin{array}{cccc|c} \boxed{1} & -1 & -1 & 1 & 0 \\ 0 & \boxed{2} & 4 & -2 & 8 \\ 0 & 0 & \boxed{0} & -1 & -4 \\ 0 & 0 & 1 & 1 & 7 \end{array}\right]$$

Unrecoverable failure

$$P_{34} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

3 & 4 row swap – permutation.

$$\left[\begin{array}{cccc|c} \boxed{1} & -1 & -1 & 1 & 0 \\ 0 & \boxed{2} & 4 & -2 & 8 \\ 0 & 0 & \boxed{1} & 1 & 7 \\ 0 & 0 & 0 & \boxed{-1} & -4 \end{array}\right]$$

$$U$$

$$P_{34} E_{32} E_{31} E_{21} A = U$$

Matrix column manipulation:

$$\begin{bmatrix} | & | & | \\ c_1 & c_2 & c_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$a \begin{bmatrix} c_1 \end{bmatrix} + b \begin{bmatrix} c_2 \end{bmatrix}$$

$$\begin{bmatrix} | & | & | \\ c_1 & c_2 & c_3 \\ | & | & | \end{bmatrix} \quad \begin{bmatrix} | & | & | \\ c_1 & c_2 & c_3 \\ | & | & | \end{bmatrix}$$

$n \times 3$ $\qquad$ $3 \times n$

$$\begin{bmatrix} | \\ c_1 \\ | \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} u & x \\ v & y \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} =$$

Row manipulation:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} u & x \\ v & y \end{bmatrix}$$

$$a[u\ x] + b[v\ y]$$

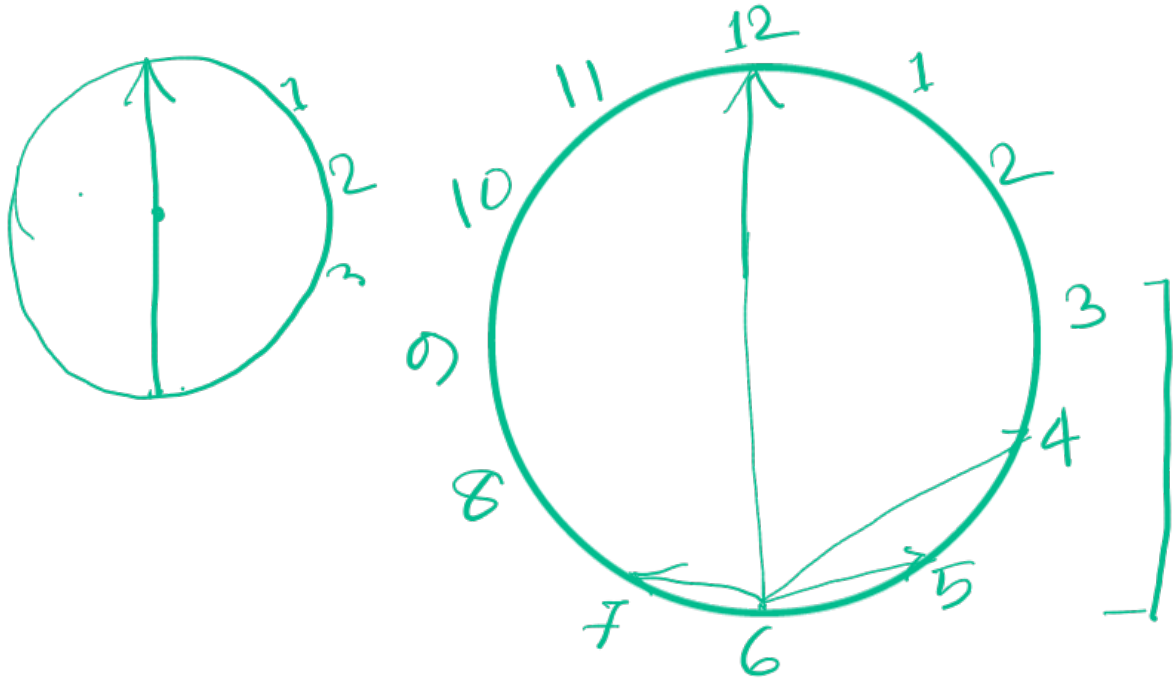
$$\begin{vmatrix} 2 & 5 & 1 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & -1 & -2 \end{vmatrix}$$

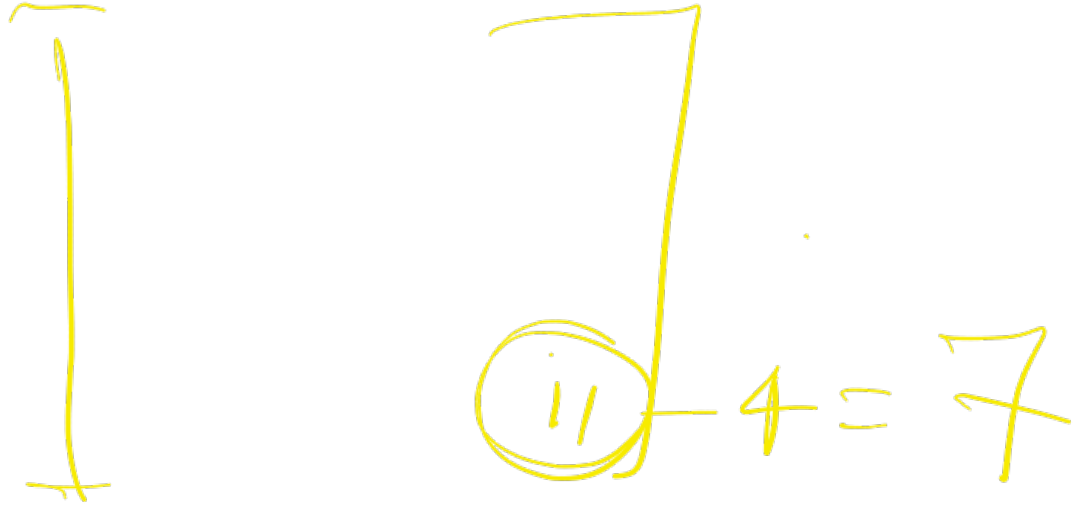
0

1 −1 3

(0,2)



A(5,5)





singular = full rank

= 5

λ_1 4



1 2 4

6 3 5

0 0 0

$\begin{bmatrix} 2 & 4 & 0 & 10 \end{bmatrix}$

$$\begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{matrix} 0 \\ 0 \\ 0 \\ 0 \end{matrix} \qquad -1/2$$

$$\Rightarrow E_{21} \begin{bmatrix} 0 & 0 & 0 & 0 \\ \frac{1}{2} & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$2 - \frac{1}{2} = \frac{3}{2}$

$-1 + 2 = 1$

$\frac{3}{2} \quad \frac{3}{2}$

$$A = \begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ 0 & 0 & \frac{3}{2} & -3 & \frac{3}{2} \\ 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{matrix} 0 \\ 0 \\ 0 \\ 0 \end{matrix} \qquad \begin{matrix} \frac{1}{2} \\ 0 \\ 1 - \frac{1}{2} \\ \frac{3}{2} \end{matrix}$$

$$E_{31} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -\frac{1}{2} & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ 0 & 0 & \frac{3}{2} & -3 & \frac{3}{2} \end{bmatrix}$$

$A =$